

BCA-5003: Computer Network

Unit I — Complete Detailed Notes + Topic-Wise PYQ Answer Book

For: BCA Semester V, BCA-5003 Computer Network

Scope: Unit I only — Basic Concepts, Data Communication, OSI Model, TCP/IP Model, addressing, topologies, and related foundation topics found in the supplied Unit-I material and PYQs.

Purpose: Long-form, exam-ready notes for high marks. Read the concept first; then learn the question-answer section attached to that concept.

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No.	Topic	PYQ focus found in supplied material
20	Final revision sheets and expected questions	—

1. Unit-I Roadmap

Unit I builds the foundation of Computer Networks. Before studying routing algorithms, LAN protocols, TCP, or application protocols in later units, you must clearly understand how two systems communicate, how data is structured layer by layer, how errors are detected, and how devices are addressed.

The best order to study

1. Learn the definitions: network, data communication, protocol, topology, error, CRC, IP, MAC.
2. Learn the five components and four characteristics of data communication.
3. Practice topology diagrams.
4. Learn error detection and CRC numerical steps.
5. Memorize OSI layers in correct order and their functions.
6. Learn TCP/IP layers and compare them with OSI.
7. Prepare the comparison tables: MAC vs IP, IPv4 vs IPv6, virtual circuit vs datagram.
8. Learn topic-wise PYQ answers exactly in the question-answer format given below.

Rule for scoring high marks

For every 8–10 mark answer, use this sequence:

1. Write a precise definition or introduction.
2. Draw a clean, labelled diagram/table.
3. Explain points under meaningful headings.
4. Add advantages, limitations, examples, or applications where suitable.
5. End with a short conclusion.

Never write only a list if the question says **explain**, **discuss**, or **describe**.

2. Computer Networks

2.1 Meaning of a Computer Network

A **computer network** is a collection of two or more autonomous devices interconnected through communication links so that they can exchange information and share resources. The devices may include computers, laptops, mobile phones, servers, printers, routers, switches, sensors, or storage systems.

The word **autonomous** means that each device can operate independently. A network does not merge computers into a single computer; rather, it connects independent systems so they can cooperate.

Networks use a combination of:

- **Hardware:** NICs, cables, fibre links, wireless access points, switches, routers, servers.
- **Software:** operating systems, network drivers, protocol implementations, security tools, applications.
- **Protocols:** agreed rules such as Ethernet, Wi-Fi, IP, TCP, UDP, DNS, and HTTP.

2.2 Basic Uses and Benefits

Resource sharing

Many users can share expensive resources such as printers, scanners, storage servers, Internet connections, licensed software, and databases.

Data sharing and communication

Networks allow users to share files and communicate using email, messaging, video conferencing, collaborative documents, and cloud applications.

Centralized administration

An organization can centrally manage user accounts, software updates, security policies, backups, and access rights.

Remote access

Authorized users can access office applications, files, cloud services, and remote desktops from another location.

Reliability and backup

Important data can be copied to centralized or distributed storage. Redundant network paths and backup systems improve availability.

Scalability

A network can grow by adding more devices, ports, switches, servers, or subnets.

2.3 Types by Geographic Coverage

Type	Full form	Approximate coverage	Example
PAN	Personal Area Network	A few metres around a person	Bluetooth headset and phone
LAN	Local Area Network	Room, home, laboratory, building	College computer lab

Type	Full form	Approximate coverage	Example
CAN	Campus Area Network	Multiple buildings in a campus	University network
MAN	Metropolitan Area Network	City or metropolitan area	City-wide service provider network
WAN	Wide Area Network	Country/continent/world	Internet, bank branch network

PYQs: Computer Networks

Q1. What is a network? Write the benefits of networks.

Asked: 2024-25

Answer

A computer network is an interconnection of two or more autonomous computers and other devices through wired or wireless communication links. It enables devices to exchange data, communicate with each other, and share hardware, software, storage, and Internet services. A network is formed using networking hardware such as network interface cards, switches, routers, cables, wireless access points, and protocols.

Benefits of computer networks

1. **Resource sharing:** A network permits multiple users to share hardware resources such as printers, scanners, storage devices, and Internet connections. This reduces unnecessary duplication and cost.
2. **File and data sharing:** Users can access shared files, databases, documents, and applications over the network. This makes collaborative work faster and more organized.
3. **Communication:** Networks support email, instant messaging, VoIP calls, video conferencing, online classes, and collaboration tools.
4. **Centralized data management:** Important files and databases can be kept on central servers. This improves control, reduces duplication, and makes data management easier.
5. **Remote access:** Authorized users can access files, applications, and systems from a remote location through VPN, remote desktop, cloud platforms, or web applications.
6. **Cost efficiency:** Sharing one printer, one Internet link, common storage, and server-based software is less expensive than providing separate resources to every computer.
7. **Better security management:** Administrators can apply authentication, authorization, firewalls, antivirus policies, backups, and access restrictions centrally.
8. **Reliability and backup:** Centralized backup and redundant systems reduce the risk of permanent data loss and improve availability.
9. **Scalability:** A network can be expanded by connecting more users, devices, switches, or branches when required.

10. **Access to information and online services:** A network connects users to websites, cloud applications, digital libraries, e-commerce, e-governance, and other Internet-based services.

Conclusion: Thus, computer networks improve communication, reduce cost, enable resource sharing, and form the foundation of modern business, education, cloud computing, and Internet services.

3. Data Communication

3.1 Definition

Data communication is the exchange of data between two or more devices through a transmission medium such as copper cable, optical fibre, radio waves, microwave, or satellite link.

The data can be text, numbers, images, audio, video, program files, control signals, or any combination of these. For successful communication, the sender and receiver must follow compatible rules, called protocols.

3.2 Components of Data Communication

A complete data-communication system has five essential components.

```
Sender  —>  Transmission Medium  —>  Receiver
|                                     |
└── Message transferred using Protocol ┘
```

Component	Explanation	Example
Message	Information being sent	Email, image, audio, file
Sender	Device that sends the message	Student's laptop
Receiver	Device that receives the message	College server
Transmission medium	Path through which signal travels	Wi-Fi, UTP cable, fibre
Protocol	Set of rules controlling communication	TCP/IP, HTTP, Ethernet

1. Message

The message is the information to be communicated. It may consist of text, numbers, pictures, sound, video, or control commands.

2. Sender

The sender is the device that originates the message. It may be a computer, workstation, smartphone, camera, terminal, or server.

3. Receiver

The receiver is the device for which the message is intended. It accepts and processes the received data.

4. Transmission Medium

The medium is the physical or wireless path through which the message travels. Examples include twisted-pair cable, coaxial cable, fibre-optic cable, Wi-Fi, radio waves, microwave, and satellite signals.

5. Protocol

A protocol is a collection of rules that governs data communication. It defines format, timing, sequencing, addressing, error handling, and interpretation of messages.

Without a common protocol, two connected devices may not understand each other. This is similar to two people speaking different languages without a translator.

3.3 Characteristics of Effective Data Communication

The effectiveness of a data-communication system depends on four characteristics.

1. Delivery

The system must deliver data to the correct destination. It should reach the intended device, user, and application—not an unauthorized or incorrect destination.

2. Accuracy

Data must reach the receiver without unwanted changes. Altered, lost, duplicated, or corrupted data is inaccurate. Error detection and correction are used to improve accuracy.

3. Timeliness

Data must arrive when it is required. Delayed data may be useless in real-time systems such as online classes, voice calls, video streaming, online gaming, medical monitoring, and industrial control.

4. Jitter

Jitter is the variation in packet-arrival time or delay. It is particularly important in real-time audio and video communication.

Example: Voice packets are expected every 20 ms. If one packet arrives after 20 ms and the next arrives after 80 ms, audio playback may become broken or uneven.

PYQs: Data Communication

Q2. What are the fundamental characteristics of effective data communication?

Asked: 2024–25

Answer

The effectiveness of a data-communication system depends on four fundamental characteristics: **delivery, accuracy, timeliness, and jitter.**

1. **Delivery:** The system must deliver data to the correct destination. The information should be received by the intended device or user and not by an unintended destination.
2. **Accuracy:** The system must deliver data accurately. Data altered during transmission because of noise, interference, or faults becomes unusable unless errors are detected and corrected.
3. **Timeliness:** The system must deliver data within an acceptable time. Late delivery may have little value, especially in real-time applications such as live video, online lectures, voice calls, and remote control systems.
4. **Jitter:** Jitter is the variation in the delay with which packets arrive at the receiver. Excessive jitter produces poor-quality audio/video, pauses, distortion, or irregular playback.

Conclusion: A communication system is effective when it sends correct data to the correct receiver at the right time with minimum variation in delay.

Q3. Write the basic components of data communication.

Asked: 2021 and 2017

Answer

Data communication is the exchange of information between two or more devices through a transmission medium. Its five basic components are:

1. **Message:** The data or information to be transmitted, such as text, image, audio, video, or a file.
2. **Sender/Transmitter:** The device that sends the message, such as a computer or smartphone.

3. **Receiver:** The device that receives the message, such as a server or another computer.
4. **Transmission Medium:** The physical or wireless path used to carry data, such as cable, fibre, radio wave, or Wi-Fi.
5. **Protocol:** A set of rules that governs communication. It specifies how data is formatted, transmitted, addressed, acknowledged, and checked for errors.

```
Message
|
Sender — Medium / Channel — Receiver
\_____ Protocol rules _____/
```

Q4. Write a short note on protocol.

Asked: 2022–23

Answer

A protocol is a set of rules and conventions that governs communication between devices in a network. It ensures that the sender and receiver interpret data in the same way.

A protocol defines:

- The format and structure of messages.
- The order in which messages are exchanged.
- Addressing rules to identify source and destination.
- Timing and synchronization rules.
- Error detection, acknowledgement, and retransmission procedures.
- Security rules where required.

For example, HTTP defines how web browsers and web servers exchange web requests and responses; TCP defines reliable transport of data; IP defines logical addressing and routing.

Without protocols, different devices may be physically connected but unable to communicate correctly.

4. Communication Modes, Line Configuration, and Transmission

4.1 Communication Modes

Communication mode refers to the direction in which data flows between two devices.

Mode	Meaning	Example
Simplex	One-way only	Keyboard to CPU, broadcast television

Mode	Meaning	Example
Half-duplex	Both directions, one at a time	Walkie-talkie
Full-duplex	Both directions simultaneously	Telephone, switched Ethernet

4.2 Line Configuration

Line configuration specifies how devices are connected to a communication link.

Point-to-point configuration

A dedicated link connects exactly two devices. The entire channel capacity is reserved for those endpoints.

Device A ————— Dedicated Link ————— Device B

Examples: a direct cable between two routers, a leased line, a point-to-point microwave link.

Multipoint configuration

More than two devices share a common link. The capacity is divided among users either by time sharing or spatial sharing.

Shared Link ———┬── Device 1
 ├── Device 2
 └── Device 3

Spatial sharing

Several devices use different portions of a link at the same time.

Time sharing

Devices take turns using the shared link.

4.3 Serial and Parallel Transmission

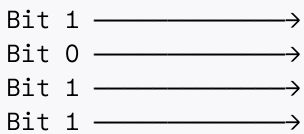
Serial transmission

Bits are sent one after another over a single communication channel. It is economical, suitable for long distances, and commonly used by modern network links.

1 → 0 → 1 → 1 → 0 → 0 → 1

Parallel transmission

Multiple bits are transmitted at the same time over multiple lines.



```
Bit 1 —————>
Bit 0 —————>
Bit 1 —————>
Bit 1 —————>
```

Parallel transmission can move a group of bits together over short distances. However, it requires more wires and faces **skew** over long distances, where signals travelling on different wires arrive at different times.

PYQs: Communication Modes and Transmission

Q5. What are the two types of line configuration? Explain.

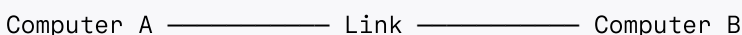
Asked: 2021

Answer

Line configuration, also called connection configuration, describes the way in which devices are connected to a communication link. A link is a physical or wireless communication path through which data travels. The two main types of line configuration are point-to-point and multipoint.

1. Point-to-point configuration

In a point-to-point configuration, a dedicated communication link connects only two devices. The complete capacity of the link is reserved for communication between those two endpoints.



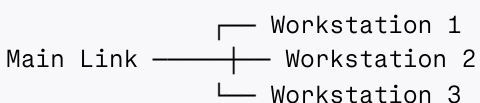
```
Computer A ————— Link ————— Computer B
```

Examples include a direct cable connection, leased telephone line, point-to-point microwave link, or router-to-router WAN link.

Advantages: dedicated bandwidth, simple management, better privacy, easier troubleshooting.

2. Multipoint configuration

In a multipoint configuration, more than two devices share a single communication link. The capacity of the channel is shared among all connected devices.



```
      Workstation 1
      |
Main Link ————+—— Workstation 2
      |
      Workstation 3
```

The sharing may be spatial, where users use separate portions simultaneously, or time-based, where users take turns.

Advantages: lower cost because a common medium is shared.

Limitation: devices must coordinate access, and performance can decrease when many devices compete for the link.

Q6. Explain the concept of parallel transmission.

Asked: 2022-23

Answer

Parallel transmission is a method of data transmission in which multiple binary digits are sent simultaneously over multiple separate communication lines. For example, if eight bits are transmitted together, eight data lines are generally required along with additional control or clock lines.

In parallel transmission, all bits of a character or data unit are sent at the same instant. This can provide high speed over a short distance. It was traditionally used inside computers, between closely placed hardware components, and in some older printer interfaces.

However, parallel transmission requires more wires and connectors, so installation cost is high. It also suffers from **timing skew** over long distances because signals on different wires may reach the receiver at slightly different times. Hence, parallel transmission is generally used for short distances, while serial transmission is preferred for long-distance communication.

5. Network Services and Distributed Processing

5.1 Network Services

A network service is a facility made available through a network to users, devices, or applications.

Directory services

Directory services store and organize information about users, groups, devices, and resources. They help in account management, login, authentication, authorization, and locating resources.

DNS

The Domain Name System translates a human-readable domain name such as `www.example.com` to an IP address required by the network.

File services

File sharing and file transfer allow users to store, access, upload, download, and exchange files through the network.

Communication services

Email, messaging, chat, voice calls, video conferencing, discussion forums, and remote desktop are common communication services.

Application services

Web applications, databases, shared storage, cloud applications, print services, and resource-sharing systems are application services.

5.2 Distributed Processing

Distributed processing is a system in which processing work is divided among multiple computers or processors connected through a network. Each device performs part of the task, and the systems coordinate to complete the overall work.

Examples include cloud computing, distributed databases, content-delivery networks, and microservice-based applications.

PYQs: Network Services and Distributed Processing

Q7. What are the two types of services used in computer networks?

Asked: 2022–23

Answer

Computer networks provide many services to users. Two broad and important categories are **directory services** and **file services**. In addition, networks commonly provide communication services and application services.

1. Directory services

Directory services organize and provide access to information about users, devices, groups, resources, and services in a network.

Important functions include:

- Storing user names and account information.

- Authentication, which verifies the identity of a user.
- Authorization, which determines what resources an authenticated user can access.
- Organizing users and systems in groups or domains.
- Name resolution through services such as DNS.

For example, DNS maps a domain name to an IP address, making Internet services easier for users to access.

2. File services

File services allow users to store, share, retrieve, and transfer files across a network.

Important file services include:

- **File sharing:** Authorized users can access common files stored on a server or shared computer.
- **File transfer:** Files can be copied or moved between systems using network protocols and applications.
- **Centralized storage:** Files can be kept on a file server or cloud storage system for backup and collaboration.

Other important network services are email, chat, remote access, web services, database services, print services, and cloud applications.

Q8. What is distributed processing in computer networks?

Asked: 2021

Answer

Distributed processing is a computing arrangement in which multiple computers or processors work together through a network to perform a common task. Instead of one central computer carrying out all operations, the processing load is divided among several systems.

Each participating computer may perform a separate part of the computation, store part of the data, or provide a specialized service. These systems communicate and coordinate using network protocols.

Advantages

1. It improves performance because several machines can work simultaneously.
2. It provides scalability, as more machines can be added when workload increases.
3. It improves reliability because failure of one machine may not stop the whole system.
4. It supports sharing of hardware, software, data, and computing power.
5. It allows services to be located near users or data sources.

Limitations

1. Coordination between machines is complex.
2. Network delay and communication failures can affect performance.
3. Security and data consistency are more difficult to manage.
4. Debugging distributed applications is harder than debugging a single-computer program.

Examples of distributed processing include cloud computing, distributed databases, online banking systems, and large web applications running on multiple servers.

6. Network Topology

6.1 Meaning

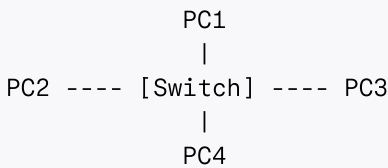
Network topology is the physical or logical arrangement of devices and communication links in a computer network. It shows how nodes are connected and how data travels between them.

- **Physical topology** describes the actual cable/device layout.
- **Logical topology** describes the logical path followed by data.

The main topologies are star, bus, ring, tree, mesh, and hybrid.

6.2 Star Topology

Every node is connected by a separate link to a central hub or, more commonly, a switch.



Advantages

- Failure of one device cable generally affects only that device.
- Easy to add or remove nodes.
- Easy fault detection and centralized control.
- Good performance with switches and dedicated full-duplex links.

Disadvantages

- Failure of the central switch can stop the network.
- More cable is required than in bus topology.
- Central equipment increases cost.

6.3 Bus Topology

All devices are attached to a common backbone cable.

```
[T] — PC1 — PC2 — PC3 — [T]
```

Advantages

- Simple and inexpensive for small legacy networks.
- Requires comparatively less cable.
- No central device is required in the basic structure.

Disadvantages

- A backbone fault can disable the whole network.
- Troubleshooting is difficult.
- Performance drops when traffic or number of devices increases.
- Traditional bus networks require terminators at cable ends.

6.4 Ring Topology

Each node connects to two neighboring nodes, forming a closed loop.

```
PC1 — PC2  
|      |  
PC4 — PC3
```

Advantages

- Controlled/ordered access in token-based designs.
- Equal opportunity for each node to transmit.
- Can give predictable performance in controlled environments.

Disadvantages

- One link/node failure can affect a simple ring.
- Adding/removing nodes can interrupt communication.
- Delay increases with more devices.

6.5 Tree Topology

A tree topology has a hierarchical arrangement of devices, often several star networks connected to a backbone or root device.

Advantages

- Easy expansion through branches.
- Supports hierarchical management.
- A problem in a lower branch may be isolated.

Disadvantages

- Failure of backbone/root device affects major portions of the network.
- Cabling and configuration become complex.
- Core devices can become performance bottlenecks.

6.6 Mesh Topology

In a mesh topology, nodes are connected through many point-to-point links.

- Full mesh: every node connects directly to every other node.
- Partial mesh: only selected nodes have multiple redundant links.

For n devices in full mesh:

$$\text{Number of links} = \frac{n(n-1)}{2}$$

Advantages

- High reliability and fault tolerance.
- Alternative paths are available.
- High privacy and dedicated links.
- A single failure usually does not stop the complete network.

Disadvantages

- Very expensive.
- Requires many cables, ports, and interfaces.
- Difficult to install and maintain for a large number of nodes.

6.7 Hybrid Topology

A hybrid topology is a combination of two or more topologies, for example star-bus or star-ring. Most large modern networks use some hybrid/hierarchical design.

PYQs: Network Topology

Q9. What is topology? Explain five types of topology with advantages and disadvantages.

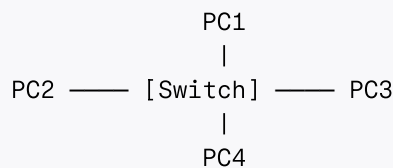
Asked: 2019

Answer

Network topology is the physical or logical arrangement of nodes and communication links in a network. It determines how devices are connected, how data flows, and how easily the network can be expanded, maintained, or repaired. The major network topologies are star, bus, ring, tree, and mesh.

1. Star topology

In star topology, all nodes are connected to a central device such as a switch.



Advantages: easy installation; easy fault isolation; failure of one cable affects only one node; simple expansion; good performance with switches.

Disadvantages: central switch failure can stop the network; more cabling is needed; central device adds cost.

2. Bus topology

In bus topology, all devices share one main backbone cable.

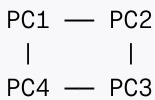


Advantages: low cost for small networks; less cable; no central device in basic form.

Disadvantages: backbone failure affects all devices; difficult fault isolation; collisions/performance issues on shared medium; limited scalability.

3. Ring topology

In ring topology, each device connects to two neighbors, forming a circular path.



Advantages: orderly access in token-based systems; equal transmission chance; predictable behavior in controlled networks.

Disadvantages: a simple ring is sensitive to link/node failures; adding/removing devices may interrupt the network; delay increases as nodes increase.

4. Tree topology

A tree topology has a hierarchical branch structure, usually based on interconnected star networks.

Advantages: highly scalable; easier administrative hierarchy; local branch faults can be isolated.

Disadvantages: dependence on root/backbone; more cabling and devices; upper devices may become bottlenecks.

5. Mesh topology

In mesh topology, nodes are connected using multiple point-to-point links. In full mesh, every node connects directly to every other node.

Advantages: highly reliable; alternate paths available; high security/privacy; failure of one link generally does not stop communication.

Disadvantages: costly; difficult installation; large cable/port requirement; complex maintenance.

Conclusion: Star topology is the most common topology in modern Ethernet LANs, while mesh is preferred where fault tolerance is critical. The choice depends on cost, scale, reliability, and performance requirements.

Q10. Differentiate network topologies used in LAN.

Asked: 2019 and 2023–24

Answer

Basis	Star	Bus	Ring	Tree	Mesh
Basic structure	All nodes connect to a central switch/hub	All nodes share one backbone	Nodes form a closed loop	Hierarchical branches	Multiple point-to-point links
Central device	Required	Not required	Not required in basic ring	Required at hierarchical levels	Not required
Failure effect	One cable failure affects one node; central failure affects all	Backbone failure affects all	Link/node failure may disrupt ring	Root/backbone failure affects many branches	One link failure usually does not stop network
Cost	Moderate	Low for small setup	Moderate	Moderate to high	Very high
Expansion	Easy	Limited	Difficult	Easy	Difficult/costly
Modern use	Very common LAN topology	Legacy/coaxial networks	Rare/legacy	Campus/enterprise hierarchy	Backbone and high-availability systems

7. Errors in Data Communication

7.1 Meaning of Error

An error occurs when the data received is not the same as the data sent. In digital communication, a transmitted bit may change from 0 to 1 or from 1 to 0 because of noise, interference, attenuation, distortion, synchronization problems, or hardware faults.

7.2 Types of Errors

Single-bit error

Only one bit changes in a data unit.

```
Sent:      1001001
Received:  1000001
```

Multiple-bit error

Two or more non-consecutive bits change.

```
Sent:      10110101
Received:  10011100
```

Burst error

Two or more bits are affected within a span of bits. The first and last bits of that span are wrong; bits in between may or may not be wrong.

Burst errors are common in real transmission channels and are an important reason for using CRC.

7.3 Error Detection and Error Correction

Error detection

The receiver determines whether received data contains an error. Common methods are parity check, checksum, and CRC.

Error correction

The receiver identifies and corrects the error itself, or the system requests retransmission of damaged data. Hamming code can correct single-bit errors; ARQ protocols correct errors through retransmission.

PYQs: Error Basics

Q11. How can errors be detected in networks? Define single-bit and multiple-bit errors.

Asked: 2018 and 2019

Answer

During data transmission, noise and other impairments can alter binary signals. An error occurs when data received by the destination differs from the data sent by the source.

Errors can be detected by adding redundant bits to the original message before transmission. The receiver uses these redundant bits to check whether the received data is valid. Common error-detection techniques are parity checking, checksum, and Cyclic Redundancy Check (CRC).

Single-bit error

A single-bit error occurs when only one bit in a data unit changes from 0 to 1 or from 1 to 0.

```
Sent:      1001001
Received:  1000001
```

Here, only one bit has changed.

Multiple-bit error

A multiple-bit error occurs when two or more bits in the data unit are altered. The changed bits do not need to be consecutive.

```
Sent:      10110101
Received:  10011100
```

Multiple-bit errors are harder to detect with simple parity. Therefore, stronger methods such as CRC are used in practical networks.

8. Error Detection: Parity, Checksum, and CRC

8.1 Parity Check

Parity is the simplest error-detection method. An additional bit called a **parity bit** is added to the data unit.

Even parity

The parity bit is chosen so that the total number of 1s, including parity, becomes even.

Odd parity

The parity bit is chosen so that the total number of 1s, including parity, becomes odd.

Limitation

Simple parity can detect an odd number of changed bits but may fail to detect an even number of bit errors. It cannot identify the error position.

8.2 Checksum

In checksum, data is divided into fixed-size words. The sender adds the words using one's-complement arithmetic and sends the complement of the sum as the checksum. The receiver repeats the operation and checks the final result.

Checksum is widely used in Internet protocols, but CRC is generally more powerful for detecting burst errors.

8.3 Cyclic Redundancy Check (CRC)

CRC is a widely used error-detection technique based on **modulo-2 division**. Modulo-2 arithmetic uses XOR operation instead of ordinary subtraction.

Main terms

Term	Meaning
Dataword	Original data bits
Generator/divisor	Predetermined bit pattern used in division
CRC bits / remainder	Redundant bits obtained after division
Codeword	Dataword followed by CRC bits

Sender-side CRC procedure

Suppose generator G has $r + 1$ bits.

1. Append r zeros to the dataword.
2. Divide the augmented dataword by G using XOR division.
3. Obtain an r -bit remainder.
4. Append the remainder to the original dataword.
5. Transmit the resulting codeword.

Receiver-side CRC procedure

1. Receiver divides the complete received codeword by the same generator.
2. If remainder is all zeros, no detectable error is found and the frame is accepted.
3. If remainder is non-zero, an error is detected and the frame is rejected/retransmitted.

Important accuracy note

A zero CRC remainder does not mathematically guarantee that no error occurred. It means no error detectable by the chosen generator was detected. In university answers, write: **“The frame is accepted when the remainder is zero; otherwise it is treated as erroneous.”**

PYQs: CRC

Q12. Write a short note on CRC.

Asked: 2025–26

Answer

CRC, or Cyclic Redundancy Check, is an error-detection technique used in data communication to check the integrity of a transmitted frame. It is based on modulo-2 binary division, in which subtraction is performed using XOR operation.

At the sender side, the original dataword is appended with zeros equal to one less than the number of bits in the generator. The augmented dataword is divided by the generator. The

remainder obtained from this division is called the CRC remainder or Frame Check Sequence. The sender appends this remainder to the original dataword and sends the resulting codeword.

At the receiver side, the received codeword is divided by the same generator. If the remainder is zero, the receiver accepts the frame because no detectable error is found. If the remainder is non-zero, the receiver concludes that the frame has been corrupted and rejects it or requests retransmission.

CRC is more reliable than simple parity checking for detecting burst errors and is widely used in data-link protocols such as Ethernet.

Q13. Explain how errors are detected using CRC.

Asked: 2023–24

Answer

CRC detects errors by appending specially calculated redundant bits to a data unit before transmission. The sender and receiver agree in advance on a generator bit pattern called the divisor.

Working at sender

1. Let the generator have $r + 1$ bits.
2. The sender appends r zeros to the dataword.
3. The augmented dataword is divided by the generator using modulo-2 division, where XOR is used instead of subtraction.
4. The remainder has r bits and is called the CRC.
5. The sender attaches this remainder to the original dataword to create the codeword.
6. The codeword is transmitted.

Working at receiver

1. The receiver receives the complete codeword.
2. It divides the codeword by the same generator using modulo-2 division.
3. If the remainder is $000 \dots 0$, the frame is accepted because no detectable error is found.
4. If the remainder is non-zero, the frame is considered damaged and is rejected or retransmitted.

Sender:

Dataword + r zeros $\rightarrow$ XOR division by generator $\rightarrow$ CRC remainder
Dataword + CRC $\rightarrow$ Codeword transmitted

Receiver:

Received codeword $\rightarrow$ XOR division by same generator

Zero remainder → Accept
Non-zero remainder → Error detected / Reject

CRC is especially effective in detecting burst errors because the generator polynomial is chosen to detect common patterns of transmission errors.

Q14. Given dataword 1001 and divisor 1011, show generation of CRC codeword at sender side.

Asked: 2021

Answer

Given:

Dataword = 1001
Divisor / Generator = 1011

The divisor has 4 bits. Therefore, the number of CRC bits required is:

$$r = 4 - 1 = 3$$

Step 1: Append zeros

Append three zeros to the dataword:

Dataword: 1001
Augmented dataword: 1001000
Generator: 1011

Step 2: Perform modulo-2 division

CRC uses XOR division. The division process is:

1001 $\oplus$ 1011 = 0010
Bring down next bit → 0100
Leading bit is 0, so XOR with 0000 → 0100
Bring down next bit → 1000
1000 $\oplus$ 1011 = 0011
Bring down next bit → 0110
Leading bit is 0, so XOR with 0000 → 0110

The last three bits are the CRC remainder:

CRC remainder = 110

Step 3: Form the codeword

Append the CRC remainder to the original dataword:

```
Codeword = Dataword + CRC
          = 1001 + 110
          = 1001110
```

Therefore, the CRC codeword transmitted by the sender is:

1001110

At the receiver, dividing 1001110 by 1011 produces remainder 000; hence the codeword passes the CRC check.

9. Hamming Code

9.1 Meaning

Hamming code is an error-correcting code that uses several parity bits placed at positions that are powers of two: 1, 2, 4, 8, and so on. It can identify and correct a single-bit error.

9.2 Parity-bit calculation

If m is the number of data bits and r is the number of parity bits, then choose r such that:

$$2^r \geq m + r + 1$$

9.3 Hamming (7,4) Code

In a 7-bit codeword, parity bits are at positions 1, 2, and 4.

Position	1	2	3	4	5	6	7
Bit role	P1	P2	D1	P4	D2	D3	D4

For even parity:

- P1 checks positions 1, 3, 5, 7.
- P2 checks positions 2, 3, 6, 7.
- P4 checks positions 4, 5, 6, 7.

The result of parity checks creates a **syndrome**. The syndrome identifies the location of the erroneous bit.

PYQ: Hamming Code

Q15. If a 7-bit Hamming codeword received is 1011011, assuming even parity, state whether it is correct or wrong.

Asked: 2017

Answer

Assume positions are numbered from left to right as 1 to 7.

Position:	1	2	3	4	5	6	7
Received:	1	0	1	1	0	1	1
Role:	P1	P2	D1	P4	D2	D3	D4

For even parity:

Check P1

P1 checks positions 1, 3, 5, and 7.

1, 1, 0, 1

Number of 1s = 3, which is odd. Therefore, P1 check fails.

Check P2

P2 checks positions 2, 3, 6, and 7.

0, 1, 1, 1

Number of 1s = 3, which is odd. Therefore, P2 check fails.

Check P4

P4 checks positions 4, 5, 6, and 7.

1, 0, 1, 1

Number of 1s = 3, which is odd. Therefore, P4 check fails.

Therefore, syndrome is:

$P4\ P2\ P1 = 111_2 = 7_{10}$

The error is at bit position 7. Flip bit 7 from 1 to 0.

Received codeword: 1011011
Correct codeword: 1011010

Conclusion: The received codeword is wrong. There is a single-bit error at position 7, and the corrected codeword is 1011010.

Note: If a question or teacher follows right-to-left bit numbering, declare that convention before solving. The method remains the same.

10. Error Control in Data-Link Layer

The data-link layer provides reliable node-to-node transfer. It uses error detection, acknowledgements, timers, sequence numbers, and retransmission to control lost or corrupted frames.

10.1 Automatic Repeat reQuest (ARQ)

ARQ is a method in which the receiver asks the sender to retransmit a frame when an error or loss occurs.

10.2 Stop-and-Wait ARQ

1. Sender sends one frame.
2. Sender waits for ACK.
3. If ACK is received before timeout, sender transmits next frame.
4. If ACK is not received or NAK/error occurs, sender retransmits the same frame.

10.3 Go-Back-N ARQ

The sender can send several frames without waiting for individual ACKs, up to a defined window size. If a particular frame is lost/damaged, sender retransmits that frame and all following frames.

10.4 Selective Repeat ARQ

The sender retransmits only the particular lost/damaged frame. Receiver can buffer correctly received out-of-order frames. It is more efficient but more complex.

PYQ: Error Control

Q16. Discuss the methods used for controlling errors in the data-link layer.

Asked: 2023–24

Answer

The data-link layer is responsible for reliable node-to-node delivery of frames. During transmission, frames may be damaged by noise, lost due to faults, duplicated because of retransmission, or received out of sequence. The data-link layer controls such errors through error detection, acknowledgements, sequence numbers, timers, and retransmission mechanisms.

1. Error detection

The sender adds redundant information to each frame. The receiver checks this information to determine whether the frame was damaged.

Common methods are:

- Parity check.
- Checksum.
- Cyclic Redundancy Check (CRC).

CRC is widely used because it detects burst errors effectively.

2. Acknowledgement and negative acknowledgement

After receiving a correct frame, the receiver sends an ACK to confirm successful receipt. In some protocols, it may send a NAK to indicate that a frame was damaged or missing.

3. Sequence numbers

Each frame is assigned a sequence number. Sequence numbers help the receiver identify missing, duplicate, and out-of-order frames.

4. Timers and retransmission

The sender starts a timer after sending a frame. If an ACK does not arrive before timeout, the sender assumes the frame or ACK was lost and retransmits the frame.

5. Stop-and-Wait ARQ

The sender transmits one frame and waits for its acknowledgement. If no acknowledgement arrives in time, the frame is retransmitted. This method is simple but inefficient on high-delay links.

6. Go-Back-N ARQ

The sender is permitted to send multiple frames before receiving acknowledgements. If an error occurs in a frame, the sender retransmits the erroneous frame and all frames sent after it in the window.

7. Selective Repeat ARQ

Only the damaged or missing frames are retransmitted. The receiver buffers correct out-of-order frames. This gives better efficiency but requires more complex logic and storage.

Conclusion: Data-link error control combines error detection with ARQ retransmission techniques to ensure reliable frame delivery across a communication link.

11. OSI Reference Model

11.1 Introduction

The **Open Systems Interconnection (OSI) reference model** is a seven-layer conceptual model developed by ISO. It organizes network communication functions into separate layers so that hardware and software from different vendors can communicate using standard interfaces and protocols.

The OSI model is mainly a reference framework. It helps students and network engineers understand where a protocol or device operates and how data moves through a network.

11.2 Seven Layers

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

Mnemonic from top to bottom:

All People Seem To Need Data Processing.

11.3 Encapsulation

At the sender, application data moves down the layers. Each layer adds control information, usually a header and sometimes a trailer. This is called **encapsulation**.

At the receiver, data moves upward and each layer removes/interprets its own control information. This is called **decapsulation**.

Layer	Common PDU
Application / Presentation / Session	Data
Transport	Segment (TCP) or datagram (UDP)
Network	Packet
Data Link	Frame
Physical	Bits

11.4 Physical Layer (Layer 1)

The physical layer transmits raw bits over a physical or wireless medium.

Functions

1. Defines electrical, optical, mechanical, and radio characteristics of the medium and interfaces.
2. Represents binary bits as signals.
3. Defines data rate or bit rate.
4. Defines connectors, pin arrangements, voltages, frequency, modulation, and line coding.
5. Defines physical topology.
6. Defines simplex, half-duplex, or full-duplex transmission mode.

Examples: UTP cable, fibre cable, connectors, radio signals, hubs, repeaters, Ethernet physical standards.

11.5 Data-Link Layer (Layer 2)

The data-link layer provides reliable transfer of frames between directly connected nodes on the same link or LAN.

Functions

1. **Framing:** divides bit stream into frames.
2. **Physical/MAC addressing:** inserts source and destination MAC addresses for local delivery.
3. **Error control:** detects damaged/lost frames and may support retransmission.
4. **Flow control:** prevents a fast sender from overloading a slow receiver on the link.
5. **Access control:** controls access to a shared medium.

Examples: Ethernet, Wi-Fi MAC, PPP, switches, bridges.

11.6 Network Layer (Layer 3)

The network layer provides source-to-destination delivery of packets across multiple networks.

Functions

1. Logical addressing through IPv4 or IPv6 addresses.
2. Routing/path selection.
3. Packet forwarding to next-hop routers.
4. Internetworking across separate networks.
5. Fragmentation/reassembly where protocol rules require it.
6. Some congestion-related functions.

Examples: IP, ICMP, routers.

11.7 Transport Layer (Layer 4)

The transport layer provides process-to-process communication and end-to-end delivery.

Functions

1. Segmentation and reassembly of application messages.
2. Port addressing and multiplexing/demultiplexing.
3. Connection control: connection-oriented or connectionless service.
4. End-to-end flow control.
5. End-to-end error control, sequencing, acknowledgement, and retransmission when supported.
6. Congestion control in TCP.

Examples: TCP and UDP.

11.8 Session Layer (Layer 5)

The session layer establishes, manages, and terminates sessions/dialogues between applications.

Functions

1. Establishes and terminates sessions.
2. Dialog control for half-duplex or full-duplex communication.
3. Adds synchronization points/checkpoints in long data transfers.
4. Supports recovery from a checkpoint after a failure.

11.9 Presentation Layer (Layer 6)

The presentation layer deals with representation, syntax, and format of data.

Functions

1. Translation/format conversion between systems.
2. Character encoding conversion, such as ASCII/Unicode representation.
3. Encryption and decryption.
4. Compression and decompression.
5. Data serialization/format handling.

11.10 Application Layer (Layer 7)

The application layer provides network services directly to end-user applications.

Functions

1. Web services and web browsing.
2. Email services.
3. File transfer and remote file access.
4. Name/directory services.
5. Remote login and network management.
6. Access to shared databases and cloud applications.

Examples: HTTP/HTTPS, DNS, SMTP, IMAP, FTP/SFTP, SSH.

PYQs: OSI Model

Q17. Draw the OSI model and explain three functions of each layer.

Asked: 2015, 2016, 2018, and 2019

Answer

The OSI reference model is a seven-layer model that standardizes network communication functions. It divides complex communication tasks into separate layers.

```
+-----+
| 7. Application Layer    |
+-----+
| 6. Presentation Layer  |
+-----+
| 5. Session Layer       |
+-----+
| 4. Transport Layer     |
```

```
+-----+
| 3. Network Layer      |
+-----+
| 2. Data-Link Layer    |
+-----+
| 1. Physical Layer     |
+-----+
```

1. Physical Layer

- Transmits raw bits over the transmission medium.
- Defines physical interfaces, connectors, signal levels, and bit rate.
- Defines physical topology and transmission mode.

2. Data-Link Layer

- Converts raw bits into frames.
- Provides MAC/physical addressing for local delivery.
- Performs error control, flow control, and medium-access control.

3. Network Layer

- Provides logical addressing using IP addresses.
- Determines routes from source network to destination network.
- Forwards packets through routers and interconnected networks.

4. Transport Layer

- Provides process-to-process delivery using port numbers.
- Segments messages and reassembles them at the destination.
- Performs end-to-end flow/error control, sequencing, acknowledgements, and retransmission where required.

5. Session Layer

- Establishes, manages, and terminates sessions between applications.
- Controls dialog between communicating systems.
- Adds synchronization points/checkpoints for recovery.

6. Presentation Layer

- Translates data formats between systems.
- Performs encryption and decryption.
- Performs compression and decompression.

7. Application Layer

- Provides services to end-user applications.
- Supports email, web, remote login, and file transfer.
- Provides access to directory, database, and network-management services.

Conclusion: The OSI model makes communication easier to understand, design, and troubleshoot by assigning different responsibilities to separate layers.

Q18. Write three characteristics/functions of the data-link layer.

Asked: 2021

Answer

The data-link layer is Layer 2 of the OSI model. It provides node-to-node delivery and converts the raw transmission capability of the physical layer into a more reliable link for the network layer.

Its main functions are:

1. **Framing:** It divides a continuous stream of bits into manageable units called frames.
2. **Physical/MAC addressing:** It adds source and destination MAC addresses to frames so that frames can be delivered to the correct device on the local network.
3. **Error control:** It detects damaged, lost, or duplicate frames using techniques such as CRC and may request retransmission.
4. **Flow control:** It controls the rate of frame transmission so that a fast sender does not overwhelm a slow receiver.
5. **Access control:** It decides which device can use a shared communication medium at a given time.

12. TCP/IP Model and Functions

12.1 Introduction

The TCP/IP protocol suite is the practical protocol family on which the Internet is built. TCP/IP is commonly represented as a four-layer model.

```
Application Layer
Transport Layer
Internet Layer
Network Access / Link Layer
```

12.2 TCP/IP Layers

1. Application Layer

This layer provides services used by network applications. It combines functions corresponding approximately to the OSI application, presentation, and session layers.

Examples: HTTP/HTTPS, DNS, SMTP, FTP, SSH, DHCP.

2. Transport Layer

This layer provides process-to-process delivery.

- **TCP:** reliable, connection-oriented protocol with sequencing, acknowledgement, retransmission, flow control, and congestion control.
- **UDP:** connectionless protocol with less overhead; it does not guarantee delivery, order, or duplicate protection.

3. Internet Layer

This layer provides logical addressing and routing across networks.

Main protocol: IP (IPv4/IPv6). Other relevant protocols include ICMP.

4. Network Access / Link Layer

This layer handles local delivery through a particular physical network, framing, and access to media.

Examples: Ethernet, Wi-Fi, PPP.

PYQ: TCP/IP

Q19. Explain the functions of TCP/IP protocol.

Asked: 2025-26

Answer

TCP/IP is a suite of communication protocols used to connect devices across local and global networks, including the Internet. It defines how data is addressed, transmitted, routed, received, and interpreted by different systems.

The major functions of TCP/IP are as follows:

1. **End-to-end communication:** TCP/IP enables devices and applications to communicate across one or more interconnected networks, from a local LAN to the global Internet.

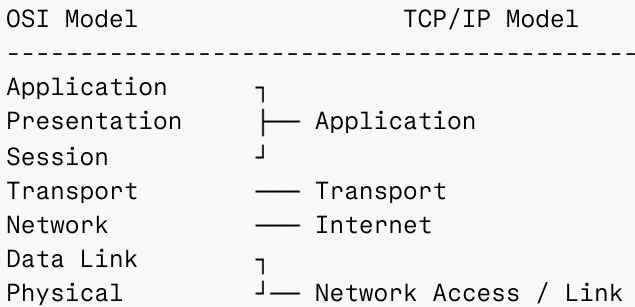
2. **Logical addressing:** The Internet Protocol assigns and uses IPv4 or IPv6 addresses to identify source and destination interfaces.
3. **Routing and forwarding:** IP packets are forwarded by routers from source network to destination network. Routers use routing information to select appropriate next hops.
4. **Process-to-process delivery:** TCP and UDP use port numbers so that incoming data reaches the correct application process, such as a web browser, email client, or DNS service.
5. **Reliable transmission through TCP:** TCP establishes a connection, sequences bytes, acknowledges received data, detects loss, retransmits missing data, and delivers data in correct order.
6. **Fast connectionless transmission through UDP:** UDP provides low-overhead datagram transfer without built-in delivery/order guarantees. It is useful for applications such as live media, DNS queries, and online games where speed can be more important than retransmission delay.
7. **Segmentation and reassembly:** Large application messages are divided into smaller transport segments for transmission and are reassembled at the destination.
8. **Flow control:** TCP prevents a fast sender from overwhelming a slow receiver by using receiver-window-based flow control.
9. **Congestion control:** TCP changes its sending behavior according to network congestion to reduce packet loss and overload.
10. **Application support:** TCP/IP supports Internet applications through protocols such as HTTP/HTTPS for the web, SMTP for email, DNS for name resolution, and FTP/SFTP for file transfer.
11. **Interoperability:** TCP/IP standards allow different devices, operating systems, vendors, and network technologies to communicate.

Conclusion: TCP/IP is the foundation of Internet communication because it combines addressing, routing, reliable or fast transport, and application-level services.

13. OSI versus TCP/IP

Conceptual Comparison

Basis	OSI Model	TCP/IP Model
Nature	Conceptual reference model	Practical Internet protocol suite/model
Developed by	ISO	U.S. DoD/DARPA Internet research community
Layers	Seven	Usually four
Upper layers	Application, Presentation, Session separate	Combined into Application layer
Lower layers	Data Link and Physical separate	Combined as Link/Network Access layer
Network service	Can conceptually support both types	IP is connectionless/best-effort
Main purpose	Standardized conceptual understanding	Actual implementation of Internet networking



14. MAC Address versus IP Address

14.1 MAC Address

MAC stands for Media Access Control. A MAC address is a link-layer identifier of a network interface. Most Ethernet and Wi-Fi MAC addresses are 48 bits and written as six hexadecimal groups.

Example:

```
00:1A:2B:3C:4D:5E
```

MAC addresses are used by switches for delivery of frames within a local network. A manufacturer usually assigns a MAC address to a NIC, but a device can often use a locally administered/spoofed MAC address through software; therefore it is incorrect to claim that MAC addresses can never change.

14.2 IP Address

An IP address is a logical network-layer address used to identify an interface and route packets across networks.

Examples:

```
IPv4: 192.168.1.10
IPv6: 2001:db8::10
```

IP addresses can be configured manually or dynamically, for example through DHCP (IPv4) or SLAAC/DHCPv6 (IPv6).

PYQ: MAC vs IP

Q20. What is the difference between MAC address and IP address?

Asked: 2024–25

Answer

Basis	MAC Address	IP Address
Full form	Media Access Control address	Internet Protocol address
OSI layer	Data-Link Layer (Layer 2)	Network Layer (Layer 3)
Main purpose	Local delivery of frames on a LAN/link	Routing packets between networks
Nature	Interface/hardware-related identifier	Logical address related to network location
Common size	Usually 48 bits	IPv4: 32 bits; IPv6: 128 bits
Representation	Hexadecimal, e.g. 00:1A:2B:3C:4D:5E	IPv4 dotted decimal; IPv6 hexadecimal colon notation
Assigned by	Usually manufacturer or local administrator	ISP, network administrator, DHCP/SLAAC/manual configuration
Changes with network	Usually more stable but may be changed/spoofed	Can change when device moves to another network or lease changes
Used by	Switches and bridges	Routers
Scope	Local network/link	Internetwork/global routing

A MAC address identifies a network interface for local frame delivery, whereas an IP address identifies an interface logically for packet delivery across networks.

How both work together

If two devices are on the same IPv4 LAN, the sender uses ARP to learn the destination device's MAC address from its IP address. It then sends an Ethernet frame to that MAC address.

If the destination is on another network, the sender usually sends the local Ethernet frame to the MAC address of the default gateway, but the IP packet still carries the remote destination's IP address. At every router hop, MAC addresses change for the local link; end-to-end IP addresses normally remain the same.

15. Responsibilities of Network Layer

The network layer is Layer 3 of the OSI model. Its main goal is source-to-destination delivery of packets across multiple interconnected networks.

Core responsibilities

1. **Logical addressing:** uses IP addresses to identify source and destination interfaces.
2. **Routing:** selects an appropriate path to reach destination networks.
3. **Forwarding:** sends a packet to the correct next-hop router/interface.
4. **Internetworking:** enables different networks to communicate.
5. **Packet handling:** handles packets and may use fragmentation/reassembly according to protocol requirements.
6. **Congestion-related support:** routers may queue, prioritize, drop, or signal packets under congestion.

PYQs: Network Layer

Q21. What are the responsibilities of the network layer?

Asked: 2024-25

Answer

The network layer is responsible for delivering packets from the source network to the destination network, even when the two devices are connected through several intermediate networks and routers.

Its major responsibilities are:

1. **Logical addressing:** The network layer assigns and uses logical addresses such as IPv4 and IPv6 addresses. These addresses identify the source and destination interfaces across networks.
2. **Routing:** The network layer determines a suitable path for packets from the source network to the destination network. Routers use routing tables and routing protocols for this purpose.
3. **Packet forwarding:** After selecting a route, a router forwards packets to the appropriate next hop.
4. **Internetworking:** It enables communication between different networks and network technologies, creating an internetwork.
5. **Fragmentation and reassembly:** When packet sizes do not match the requirements of a network link, fragmentation may be needed according to protocol rules. Reassembly reconstructs the original packet where applicable.
6. **Traffic and congestion handling:** Network devices manage packets during congestion through buffering, queueing, prioritization, dropping, and control mechanisms.

Conclusion: The network layer makes end-to-end packet delivery across multiple networks possible through IP addressing, routing, and forwarding.

16. IPv4 Addressing

16.1 Definition

IPv4 stands for Internet Protocol version 4. It uses a 32-bit logical address and is commonly written in dotted-decimal notation.

```
192.168.1.1
```

A 32-bit IPv4 address gives a theoretical address space of:

$$2^{32} = 4,294,967,296$$

Not every address is available for public hosts because several ranges are reserved for private use, multicast, loopback, documentation, and other functions.

16.2 Prefix and Host Portion

Modern IPv4 uses CIDR prefix notation.

```
192.168.1.0/24
```

/24 means that the first 24 bits form the network prefix. The remaining bits are used for host/interface addressing in that network.

16.3 Historical Classful Addressing

Classful addressing is historically important for exams, but modern networks use CIDR.

Class	First octet range	Default mask	Historical role
A	1–126	/8	Very large networks
B	128–191	/16	Medium networks
C	192–223	/24	Small networks
D	224–239	Not applicable	Multicast
E	240–255	Not applicable	Experimental/reserved

16.4 IPv4 Limitations

- 32-bit address space is limited.
- Public address exhaustion led to widespread NAT use.
- Header is comparatively more complex than IPv6 base header.
- Security is not automatically present; IPsec support is optional in IPv4 deployments.

PYQ: IPv4

Q22. Write a short note on IPv4 addressing.

Asked: 2024-25

Answer

IPv4, or Internet Protocol version 4, is the fourth version of the Internet Protocol. It provides logical addressing and routing for devices communicating across interconnected networks. IPv4 uses a 32-bit address, allowing a theoretical maximum of 2^{32} , or approximately 4.29 billion, unique address values.

An IPv4 address is written in dotted-decimal notation as four 8-bit octets. For example:

```
192.168.1.1
```

Each IPv4 address contains a network part and a host/interface part. Modern networks use CIDR notation, such as 192.168.1.0/24, where /24 identifies the length of the network prefix.

Historically, IPv4 addresses were divided into classes A, B, C, D, and E. Today, classless addressing or CIDR is used because it allocates addresses more efficiently.

The major limitation of IPv4 is address exhaustion due to the growth of Internet-connected devices. Techniques such as subnetting, CIDR, DHCP, and NAT help conserve address space. IPv4 remains widely used, but IPv6 has been developed to provide a much larger address space and improved Internet-layer features.

17. IPv6 Addressing

17.1 Definition

IPv6 stands for Internet Protocol version 6. It uses 128-bit addresses and was designed to overcome IPv4 address exhaustion.

Example:

```
2001:0db8:0000:0000:0000:0000:0001
```

Compressed form:

```
2001:db8::1
```

17.2 IPv6 Notation Rules

1. An IPv6 address contains eight groups of four hexadecimal digits.
2. Leading zeros in a group may be omitted.
3. One consecutive sequence of zero groups may be shortened using `::`.
4. `::` can be used only once in an address.

17.3 Features

- 128-bit, very large address space.
- Simplified fixed-size base header.
- Extension headers for optional features.
- Auto-configuration using SLAAC and possible DHCPv6.
- Multicast and anycast; no broadcast.
- Better hierarchical routing support.
- IPsec support is standardized, but IPv6 traffic is not automatically encrypted in every use case.

PYQ: IPv6

Q23. Write a short note on IPv6 addressing.

Asked: 2024–25

Answer

IPv6, or Internet Protocol version 6, is the successor to IPv4. It was developed mainly to solve the shortage of IPv4 addresses and to provide improved Internet-layer operation. IPv6 uses a 128-bit address, which gives an extremely large address space of 2^{128} possible addresses.

IPv6 addresses are written in hexadecimal notation using colon-separated groups. For example:

```
2001:0db8:0000:0000:0000:0000:0000:0001
```

The same address can be shortened as:

```
2001:db8::1
```

IPv6 allows leading zeros to be omitted and one consecutive sequence of zero groups to be replaced by `::`.

Important features of IPv6 include a very large address space, simplified base header, extension headers, efficient hierarchical routing, multicast and anycast communication, and

automatic address configuration through SLAAC. IPv6 does not use broadcast; multicast is used instead. IPv6 supports IPsec standards, although encryption is not automatically enabled for every IPv6 packet.

IPv6 reduces the need for NAT solely for address conservation and supports direct end-to-end addressing more easily than IPv4. It is increasingly important as the number of Internet-connected devices grows.

18. Virtual Circuits and Datagram Networks

18.1 Virtual Circuit

A virtual circuit is a logical connection established between sender and receiver before actual data transfer. Packets generally follow the same predetermined path, and network devices maintain information about the connection.

Phases:

1. Connection setup.
2. Data transfer.
3. Connection release.

Examples: X.25, Frame Relay, ATM.

18.2 Datagram Network

A datagram network is connectionless. Each packet is treated independently and can be routed through a different path. Packets may arrive out of order, be delayed differently, or be lost.

The Internet Protocol is the standard example of a datagram network.

PYQs: Virtual Circuit and Datagram

Q24. Compare virtual circuits and datagrams.

Asked: 2023–24

Answer

Basis	Virtual Circuit Network	Datagram Network
Connection setup	Required before data transfer	Not required
Nature	Connection-oriented	Connectionless
Routing	Path decided during setup and normally followed by all packets	Each packet is routed independently

Basis	Virtual Circuit Network	Datagram Network
Packet order	Usually preserved	May arrive out of order
Packet header	Often contains a short virtual-circuit identifier	Contains complete destination addressing information
Router state	Network devices maintain connection state	Basic routers generally do not maintain per-connection state
Failure handling	Circuit may need to be re-established	Later packets can use another route
Overhead	Setup overhead, then smaller per-packet route information	No setup overhead, but more addressing information per packet
Example	X.25, Frame Relay, ATM	IP/Internet

A virtual-circuit network is appropriate when an established logical path and orderly delivery are desired. A datagram network provides flexibility because every packet can be routed independently.

Q25. Explain virtual circuit subnet.

Asked: 2021

Answer

A virtual-circuit subnet is a communication network in which a logical path, called a virtual circuit, is established between the sender and receiver before data transmission begins. It is called virtual because no dedicated physical wire is reserved exclusively for the two users; instead, a logical connection is created over shared physical network resources.

The operation has three phases:

1. **Setup phase:** The network establishes a route and creates a virtual-circuit identifier in the relevant switches/routers.
2. **Data-transfer phase:** All packets follow the same established path and carry the virtual-circuit identifier.
3. **Release phase:** The connection is terminated and network resources/state are released.

Virtual circuits can maintain packet order because all packets normally follow one path. They also make traffic management and quality-related control easier. Examples include X.25, Frame Relay, and ATM.

19. DTE–DCE Interface

19.1 DTE

DTE stands for **Data Terminal Equipment**. It is a device that produces or consumes user data.

Examples: computer, terminal, router interface in a serial setup.

19.2 DCE

DCE stands for **Data Circuit-terminating Equipment**. It provides the connection between DTE and the transmission medium and may perform signal conversion, timing/clocking, and line interfacing.

Examples: modem, CSU/DSU, communication equipment provided by a network service provider.

```
Computer (DTE) → Modem (DCE) → Communication Link → Modem (DCE) → Computer (DTE)
```

PYQ: DTE–DCE

Q26. Explain DTE–DCE interface.

Asked: 2021

Answer

DTE and DCE are terms used in data communication, especially in serial and WAN communication.

DTE (Data Terminal Equipment) is the device that acts as a source or destination of digital user data. A computer, terminal, or router is a common example of DTE.

DCE (Data Circuit-terminating Equipment) is the device that provides the interface between DTE and the communication medium. It may convert signals, provide timing/clocking, and place data on the transmission line. A modem is a common example of DCE.

```
DTE          DCE          Medium          DCE
Computer ——— Modem/CSU-DSU ——— Transmission Link ——— Modem/CSU-DSU ——— Cor
```

At the sender side, DTE gives digital data to DCE. The DCE converts it into signals suitable for the medium and transmits it. At the receiving side, DCE receives the signal, converts it into usable digital data, and gives it to DTE.

Thus, DTE produces/uses data, while DCE connects DTE to the communication line.

20. Final Revision Sheet

20.1 Definitions to Memorize

Term	One-line definition
Computer network	Interconnection of autonomous devices for sharing data, resources, and services
Data communication	Exchange of data between devices through a transmission medium
Protocol	Rules governing communication between devices
Jitter	Variation in packet-arrival delay
Topology	Physical or logical arrangement of nodes and links
CRC	Error-detection method based on modulo-2/XOR division
MAC address	Layer-2 identifier used for local frame delivery
IP address	Layer-3 logical address used for routing across networks
DTE	Device that produces or consumes user data
DCE	Device that interfaces DTE with transmission medium
Virtual circuit	Pre-established logical connection for data transfer
Datagram	Independently routed connectionless packet

20.2 OSI Layers: Rapid Recall

7 Application → User services: web, email, file transfer
6 Presentation → Translation, encryption, compression
5 Session → Dialog control, synchronization
4 Transport → Ports, segmentation, reliable delivery
3 Network → IP addressing, routing, packets
2 Data Link → Frames, MAC, link error/flow control
1 Physical → Bits, signals, cable, connectors

20.3 High-Probability Questions

1. Explain OSI model with functions of every layer.
2. Explain CRC and solve a CRC numerical.
3. Explain data communication components and characteristics.
4. Explain network topologies with advantages/disadvantages.
5. Compare MAC address and IP address.
6. Write notes on IPv4 and IPv6.
7. Explain TCP/IP functions and compare TCP/IP with OSI.
8. Explain error control in data-link layer.
9. Compare virtual circuit and datagram networks.

10. Explain DTE–DCE and point-to-point/multipoint configuration.

20.4 Last-Minute Corrections

- CRC uses **XOR/modulo-2 division**, not normal decimal division.
- A zero CRC remainder means **no detectable error**, not certainty of perfect data.
- TCP provides reliability; UDP does not guarantee delivery or order.
- Routers use IP addresses at Layer 3; switches use MAC addresses at Layer 2.
- ARP maps an IPv4 address to MAC address on a local network. IPv6 uses Neighbor Discovery, not ARP.
- MAC addresses can commonly be locally administered/spoofed; do not say they can never change.
- IPv6 does not use broadcast; it uses multicast/anycast.

21. 7-Day Unit-I 10-CGPA Plan

Day	Main target	Practice output
1	Network basics, data communication, protocol, characteristics	Write Q1–Q4 without looking
2	Modes, line configuration, services, distributed processing	Draw point-to-point/multipoint diagrams; write Q5–Q8
3	Topology	Draw all topology diagrams and learn Q9–Q10
4	Errors, parity, CRC	Solve CRC numerical at least five times; write Q11–Q14
5	Hamming and ARQ	Solve one Hamming question and write Q15–Q16
6	OSI and TCP/IP	Draw OSI/TCP-IP mapping; write Q17–Q19
7	Addressing, virtual circuit/datagram, DTE–DCE	Revise Q20–Q26 and give one timed mock test

Final Instruction

For 10-CGPA-level preparation, do not only read. Use active recall: close the notes, draw diagrams from memory, write each PYQ answer in your own handwriting, compare with this guide, and rewrite weak answers. Repeat CRC, OSI layers, topology diagrams, and comparison tables until you can reproduce them without prompts.

Prepared from the supplied Unit-I notes, syllabus information, and available BCA-5003 PYQ indicators. Technical statements have been corrected for accuracy and exam-safe wording.